Neural Networks NERC
General Structure
Detailed Structure
Core task
Goals & Deliverables

Advanced Human Language Technologies

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Outline

1 Neural Networks NERC

2 General Structure

3 Detailed Structure
- Learner
- Classifier
- Auxiliary classes

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Task 1.2 - NERC using neural networks

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Assignment

Write a python program that parses the given XML and recognizes and classifies drug names. The program must use a neural network approach.

```
$ python3 ./nn-NER.py devel.xml result.out
DDI-DrugBank.d278.s0|0-9|Enoxaparin|drug
DDI-DrugBank.d278.s0|93-108|pharmacokinetics|group
DDI-DrugBank.d278.s0|113-124|eptifibatide|drug
DDI-MedLine.d88.s0|15-30|chlordiazepoxide|drug
DDI-MedLine.d88.s0|33-43|amphetamine|drug
DDI-MedLine.d88.s0|49-55|cocaine|drug
DDI-MedLine.d88.s1|82-95|benzodiazepine|drug
...
```

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General Structure

The general structure is basically the same than for the traditional ML approach:

- B-I-O schema
- Two programs: one learner and one classifier.
- The learner loads the training (Train) and validation (Devel) data, formats/encodes it appropiately, and feeds it to the model, toghether with the ground truth.
- The classifier loads the test data, formats/encodes it in the same way that was used in training, and feeds it to the model to get a prediction.

In the case of NN, we don't need to extract features (though we do need proper input encoding)

normalmente tienes 5000 neuronas y un vector de input de 5000 de manera que solo 1 es 1 y el resto 0

we need to encode the vector, but we do not need feature

Input Encoding

- The input/output layers of a NN are vectors of neurons, each set to 0/1.
- Modern deep learning libraries handle this in the form of *indexes* (i.e. just provided the *position* of active neurons, ommitting zeros). si es 26 significa que es todo 0 menos en el indice 26
- For instance, in a LSTM, each input word in the sequence may be encoded as the concatenation of different vectors each containing information about some aspect of the word (form, lemma, PoS, suffix...)
- Each vector will have only one active neuron (*one-hot-encoding*), indicated by its *index*. This input is usually fed to an embedding layer.
- Our learner will need to create and store *index* dictionaries to be able to map the index number assigned to each word, label, or any other used piece of information. See class *Codemaps*.

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Learner - Main program

```
def train(trainfile, validationfile, params, modelname) :
   '''
   learns a NN model using trainfile as training data, and validationfile
   as validation data. Saves learnt model in a file named modelname
   '''
   # load train and validation data in a suitable form
   traindata = Dataset(traindir)
   valdata = Dataset(validationdir)

   # create indexes from training data
   codes = Codemaps(traindata, params)
   # encode datasets
   train_loader = encode_dataset(traindata, codes, params)
   val_loader = encode_dataset(valdata, codes, params)

   # build network
   network = nercLSTM(codes)

   # save indexs
   os.makedirs(modelname,exist_ok=True)
   codes.save(modelname+"/codemaps")

   # train network, keep the best performing model
   best = 0
   for epoch in range(params['epochs']):
      acc = train(network, epoch, train_loader)
      if acc>best :
         best = acc
         torch.save(network, os.path.join(modelname,f"network.nn"))
```

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Classifier - Main program

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```
def predict(modelname, datafile, params, outfile) :
    '''
    Loads a NN model from 'modelname' and uses it to extract drugs
    in datafile. Saves results to 'outfile' in the appropriate format.
    '''
    # Load model
    model = torch.load(os.path.join(modelname,"network.nn"),
                       map_location=torch.device(used_device))
    model.eval()
    # load indexes
    codes = Codemaps(os.path.join(modelname,"codemaps"), params)
    # load data to annotate
    testdata = Dataset(datafile)
    test_loader = encode_dataset(testdata, codes, params)
    # run each sentence through the NN, get results
    Y = []
    for X in test_loader:
        y = model.forward(*X)
        Y.extend([[codes.idx2label(torch.argmax(w)) for w in s] for s in y] )

    # print results
    output_entities(testdata, Y, codes, outfile)
```

you have for every sentence in the batch o dataset and for each word in a sentence

translates back the output

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Auxiliary classes - Dataset

```
class Dataset:
   ## constructor: parses datafile XML file, tokenizes each sentence, and
   ## stores a list of sentences, as well as ground truth for each token
   def __init__(self, datafile)

   ## iterator to get all sentences in the data set.
   ## for each sentence returns a triplet (text, tokens, labels)
   def sentences(self)

   ## iterator to get ids for sentence in the data set
   def sentence_ids(self)

   ##  get tokens for one  given its id
   def get_sentence_tokens(self, sid) :
   ##  get text for one sentence given its id
   def get_sentence_text(self, sid) :
   ##  get labels for one sentence given its id
   def get_sentence_labels(self, sid) :
  '''
```

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Auxiliary classes - Codemaps

```
class Codemaps :
    # Constructor: create code mapper either from training data, or
    #              loading codemaps from given file.
    #              If 'data' is a Dataset, and lengths are not None,
    #              create maps from given data.
    #              If data is a string (file name), load maps from file.
    def __init__(self, data, params)
    # Save created codemaps in file named 'name'
    def save(self, name)
    # Convert a Dataset into lists of word codes and sufix codes
    # Adds padding and unknown word codes.
    def encode_words(self, data)
    # Convert the gold labels in given Dataset into a list of label codes.
    # Adds padding
    def encode_labels(self, data)
    # get word index size
    def get_n_words(self)
    # get suf index size
    def get_n_sufs(self)
    # get label index size
    def get_n_labels(self)
    # get index for given word
    def word2idx(self, w)
    # get index for given suffix
    def suff2idx(self, s)
    # get index for given label
    def label2idx(self, l)
    # get label name for given index
    def idx2label(self, i)
```

handles the translation

Required functions - network.py

podemos modificar los sizes dels embeddings

```
class nercLSTM(nn.Module):
   def __init__(self, codes):
      super(nercLSTM, self).__init__()
      # get sizes from index
      n_words = codes.get_n_words()
      n_sufs = codes.get_n_sufs()
      n_labels = codes.get_n_labels()
      # create embedding layers
      embW_sz, embS_sz = 100, 50
```

we have the embeddings for the words and the suffixes

```
      self.embW = nn.Embedding(n_words, embW_sz)
      self.embS = nn.Embedding(n_sufs, embS_sz)
      self.dropW = nn.Dropout(0.1)
      self.dropS = nn.Dropout(0.1)
      # create LSTM layer + final linear classification layer
      lstm_in_sz, lstm_out_sz = embW_sz+embS_sz, 200
      self.lstm = nn.LSTM(lstm_in_sz, lstm_out_sz,
                          bidirectional=True, batch_first=True)
      self.out = nn.Linear(2*lstm_out_sz, n_labels)
```

since the lstm is bidirectional the output is twice the size

concatenates and links the layers

```
   def forward(self, w, s):
      x = self.embW(w)  # apply embedding layers to input
      y = self.embS(s)
      x = self.dropW(x) # apply dropout to embeddings output
      y = self.dropS(y)
      # concatenate embeddigns for word + suffix
      x = torch.cat((x, y), dim=2)
      x = self.lstm(x)[0]  # feed concatenated vector to LSTM
      x = self.out(x)  # feed LSTM output to classification layer
      return x
```

Network architecture

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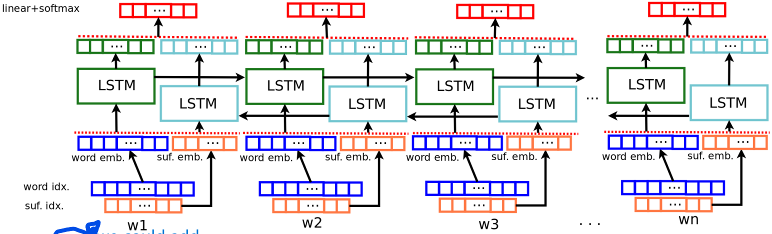

we could add more vectors f.i. for the pos, lemmas...

añadir un vector de features sin embedding (ver siguiente)

the idea is that we can extend the network or try with different

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play with the architecture of the network

Build a good NN-based drug NERC

Strategy: Experiment with different architectures and possibilities. Some elements you can play with:

- Embedding layers dimensions
- Initialitzing word embeddings with available pretrained models
- Max length and suffix length values
- Number and/or size of LSTM layers
- Used optimizer, learning rate, batch size, ...
- Number and kind of layers or activation functions
- Additional input layers (maybe with embeddings). Attention: This will require extending class `Codemaps` to handle the codes of added input layers.
    - different length suffixes and/or prefixes
    - PoS tags
    - lemmas
    - feature layer (with information about capitalization, dashes, presence in external resources, etc)

una layer que tenga nuestras features, la network no sabe como son las palabras si estan en mayusculas o con slash. No lo tenemos que embed xq ya son vectores de 1 y 0

Build a good NN-based drug NERC

Warnings:

- Neural Network training uses randomization, so different runs of the same program will produce different results. For repeatable results, use a random seed (and/or run the training several times).
- During training, *accuracy* on training and validation sets is reported. Those values are usually over 98%. However, this is due to the fact that most of the words have label "O" (non-drug). Accuracy values around 98% roughly correspond to F_1 values under 25%. To get a reasonable F_1, validation set accuracy should reach about 99.5%.

  To precisely evaluate how your model is doing, do not rely on reported accuracy: run the classifier on the development set and use the evaluator.

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Exercise Goals

What you should do:

- Experiment with architecture variations.
- Experiment with different learning hyperparameters.
- Experiment with different input information
- Keep track of tried variants and parameter combinations.

What you should **NOT** do:

- Get insights about errors from `devel` dataset. You have `train` for that. `devel` is only used to evaluate the performance of a given configuration.
- Select architectures or hyperparameters based on system performance on `test` dataset. You have `devel` for that. `test` is only used to evaluate model generalization ability once the best configuration has been chosen.

Deliverables

PYTHON3 RUN.PY PARSE
PYTHON3 RUN.PY TRAIN
EN RUN.PY HAY INSTRUCCIONES

At the end of the NERC task, you will need to deliver a single report on the work carried out on NERC-ML, NERC-NN, and NERC-LLM systems

So, during the development and experimentation on NERC-NN:

- keep track of tried/discarded architectures/hyperparameters
- keep track of tried/discarded input information.
- Record obtained results in the different experiments, and compile the information you'll later need to elaborate the report.

one of the goals is comparing nn with classical ml, was it faster?? ...